

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fourth Semester B.E. (E&E) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EE404/20EE503 – LINEAR CONTROL SYSTEMS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I

1. a) Compare open loop and closed loop systems with examples.
- b) For the system shown in Fig. 1b. Obtain:
 - i. Mechanical network
 - ii. Differential equations governing the mechanical system

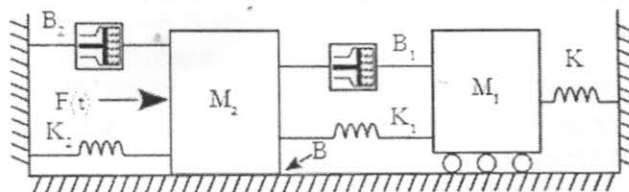


Figure 1(b)

- c) Using block diagram reduction techniques, find the transfer function for the figure shown in Fig. 1(c)

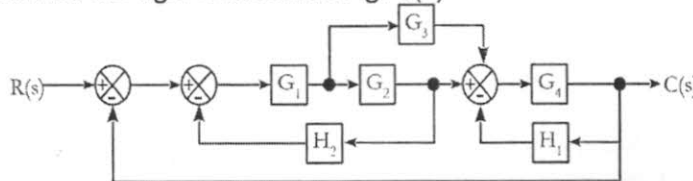


Figure 1(c)

2. a) For the system shown in Fig. 2(a). Obtain:
 - i. Differential equations governing the mechanical system
 - ii. Torque current analogy

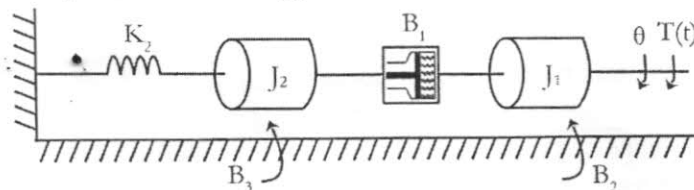


Figure 2(a)

- b) Obtain the transfer function of the gear train system shown in Figure. 2(b). referred to Gear-1.

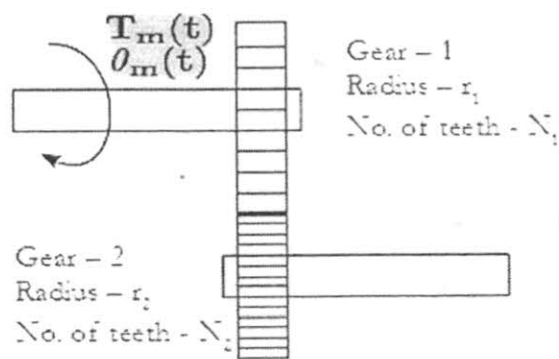


Figure 2(b)

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- c) For the system represented by the following equations, find the transfer function $X(s)/U(s)$ by signal flow graph technique

$$\begin{aligned}x &= x_1 + b_3 u \\ \dot{x}_1 &= -a_1 x_1 + x_2 + b_2 u \\ \dot{x}_2 &= -a_2 x_1 + b_1 u\end{aligned}$$

08 L2 2 2

3. a) A unity feedback system has $G(s) = \frac{K}{s(s+2)(s^2+2s+5)}$. Find K so as to have steady state error less than $e_{ss} < 0.2$ for unit ramp input. Determine e_{ss} if input is $r(t) = 2 + 4t + \frac{t^2}{2}$
- b) Obtain the transfer function between output displacement and armature applied voltage for an armature controlled DC motor.
- c) Derive the expressions for the following time response specifications for an underdamped second order system
- Rise time
 - Peak time
 - Peak overshoot

06 L3 2 2

06 L2 1 2

08 L2 2 2

Unit - II

4. a) Explain the BIBO stability for an LTI system
- b) By means of the Routh criterion determine the stability of the systems represented by the characteristic equation $s^6 + 3s^5 + 5s^4 + 9s^3 + 8s^2 + 6s + 4 = 0$
- c) Obtain the expression for the resonant frequency and resonance peak of a second order under damped system.

06 L1 3 4

4

06 L3 3 4

08 L2 4 4

5. a) A feedback control system has an open-loop transfer function

$$G(s)H(s) = \frac{K}{s(s+3)(s^2+2s+2)}$$

Draw the root locus diagram as K is varied from 0 to ∞ .

10 L3 3 4

- b) Sketch the bode plot for the given transfer function. Determine gain margin, phase margin.

$$G(s) = \frac{10}{s(1+0.5s)(1+0.01s)}$$

10 L4 4 4

6. a) A feedback control system has an open-loop transfer function

$$G(s)H(s) = \frac{K}{(s+1)(s+3)(s+6)}$$

Draw the root locus plot as K is varied from 0 to ∞ . Also find the range of K for stability.

10 L3 3 4

- b) Find the transfer function for the given the bode plot in Figure 6b

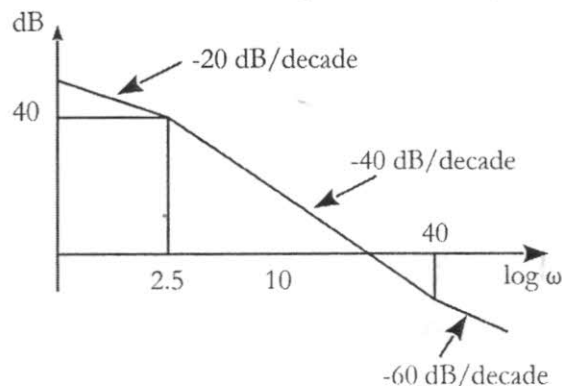


Figure. 6b

-2-

10 L3 4 4

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Unit – III

7. a) Draw the RC lag compensating network. With the help of frequency response illustrate its characteristics.
b) Explain PI controllers.
c) Obtain state equation and output equation for the electrical network shown in figure 7(c)

6	L1	5	4
6	L2	5	4

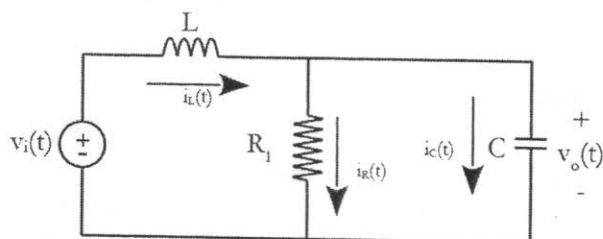


Figure. 7(c)

8	L3	5	4
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8. a) Draw the RC lead compensating network. With the help of frequency response, illustrate its characteristics.
b) Explain PID controllers.
c) Obtain state equation and output equation for the electrical network shown in figure 8(c)

6	L1	5	4
6	L2	5	4

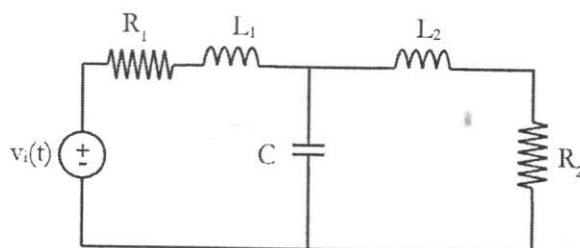


Figure. 8(c)

8	L3	5	4
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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
